

79. Apple Authentication 3.0 Coprocessor

The Apple Authentication 3.0 Coprocessor is an accessory hardware component containing an X.509 certificate enabling the generation of responses to authentication challenges.

79.1 Overview

Apple Authentication 3.0 Coprocessor (Auth 3.0 CP) is used to authenticate an accessory with devices running iOS 10.0.2 or later. See [Accessory Authentication](#) (page 276) for more information.

The Auth 3.0 CP is available in the following versions:

- [XDFN Package](#) (page 696).
- [WLCSP Package](#) (page 698).

See [Table 100-1](#) (page 868) for part numbers.

The Auth 3.0 CP Export Classification Control Number (ECCN) is EAR99. Both XDFN and WLCSP versions have a moisture sensitivity level (MSL) of 1.

79.2 Mechanical

This section provides technical details and tolerances for the Auth 3.0 CP chip.

79.2.1 XDFN Package

[Figure 79-1](#) (page 697) shows the Auth 3.0 CP's package's layout, pad locations, and dimensional tolerances.

Figure 79-1 Auth 3.0 CP package (XDFN)

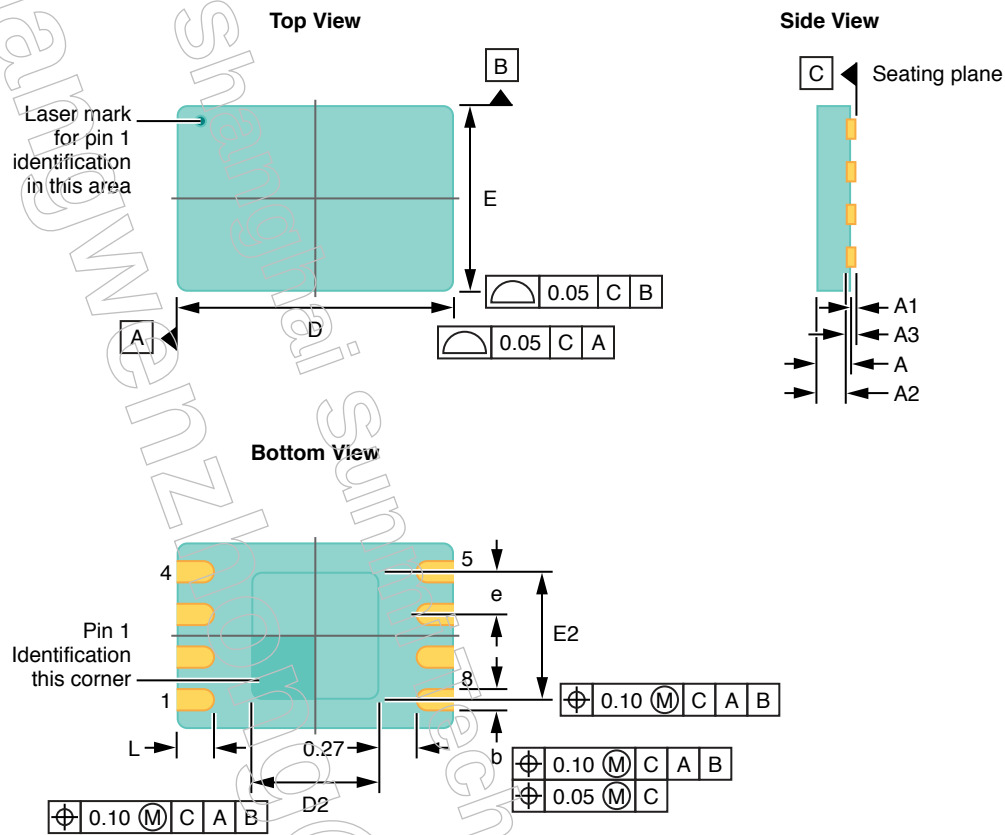


Table 79-1 Auth 3.0 CP dimensions (XDFN)

Symbol	Minimum	Nominal	Maximum
A			0.40 mm
A1			0.05 mm
A2		0.23 mm	0.28 mm
A3		0.127 mm (REF)	
b	0.18 mm	0.25 mm	0.30 mm
D		3.00 mm (BSC)	
D2	1.56 mm	1.66 mm	1.76 mm
E		2.00 mm (BSC)	
E2	1.40 mm	1.50 mm	1.60 mm
L	0.35 mm	0.40 mm	0.45 mm
e		0.50 mm (BSC)	

Figure 79-2 (page 698) and Table 79-2 (page 698) show the Auth 3.0 CP's recommended land pattern dimensions. The package includes a central paddle (exposed pad) which may be soldered but left Not Connected (NC) or may be connected to Ground.

Figure 79-2 Auth 3.0 CP land pattern dimensions (XDFN)

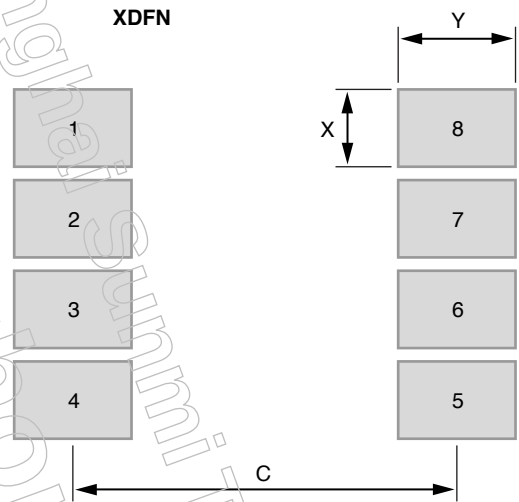


Table 79-2 Auth 3.0 CP land pattern dimensions (XDFN)

Dimension/Symbol	Value
Body	2 mm x 3 mm
Pitch	0.50 mm
Lead Length	0.45 mm
C	2.90 mm
X	0.30 mm
Y	0.90 mm

79.2.2 WLCSP Package

Figure 79-3 (page 699) shows the Auth 3.0 CP's package's layout, pad locations, and dimensional tolerances.

Figure 79-3 Auth 3.0 CP package (WLCSP)

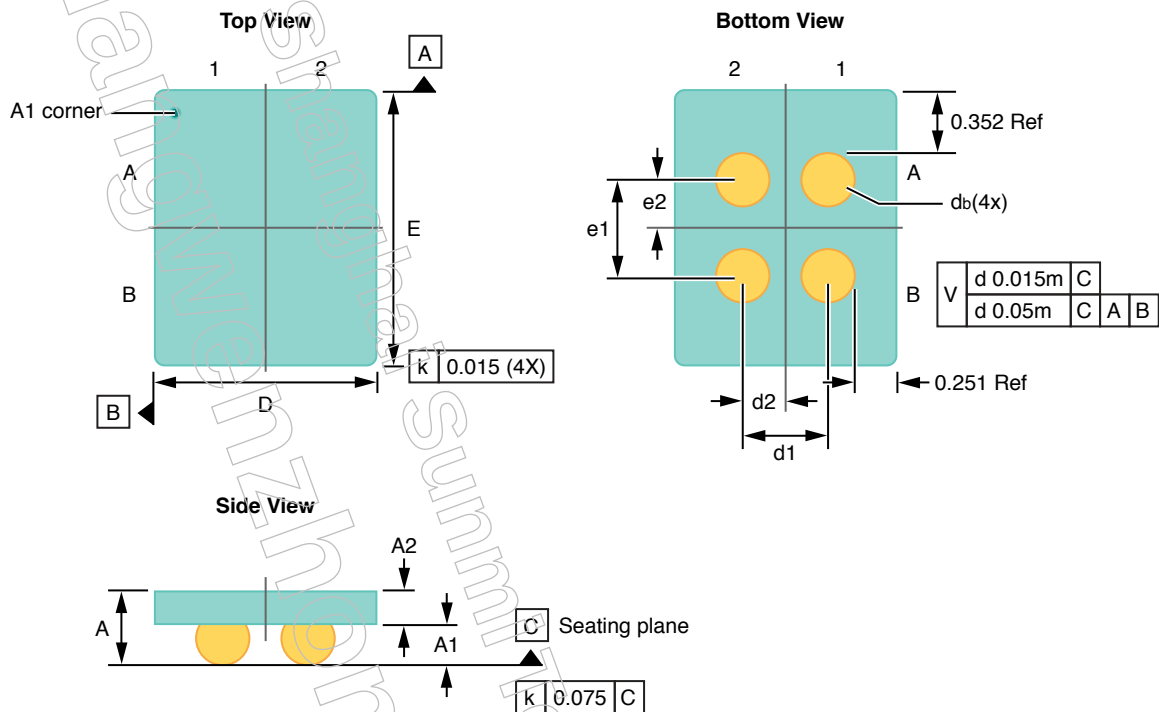


Table 79-3 Auth 3.0 CP dimensions (WLCSP)

Symbol	Minimum	Nominal	Maximum
A	0.330 mm	0.360 mm	0.390 mm
A1	0.160 mm	0.175 mm	0.190 mm
A2	0.170 mm	0.185 mm	0.200 mm
D	1.122 mm	1.152 mm	1.182 mm
d1	0.400 mm (BSC)		
d2	0.200 mm (BSC)		
E	1.424 mm	1.454 mm	1.484 mm
e1	0.500 mm (BSC)		
e2	0.250 mm (BSC)		
b	0.230 mm	0.250 mm	0.270 mm

The ball shear is 161 g/ball. Apple recommends mounting Auth 3.0 CP on a PCB with 227 μm diameter round pads.

79.2.3 Packing

Figure 79-4 (page 700) and Figure 79-5 (page 701) describe how the Auth 3.0 CP is delivered to accessory manufacturing facilities.

Figure 79-4 Auth 3.0 CP packing tape

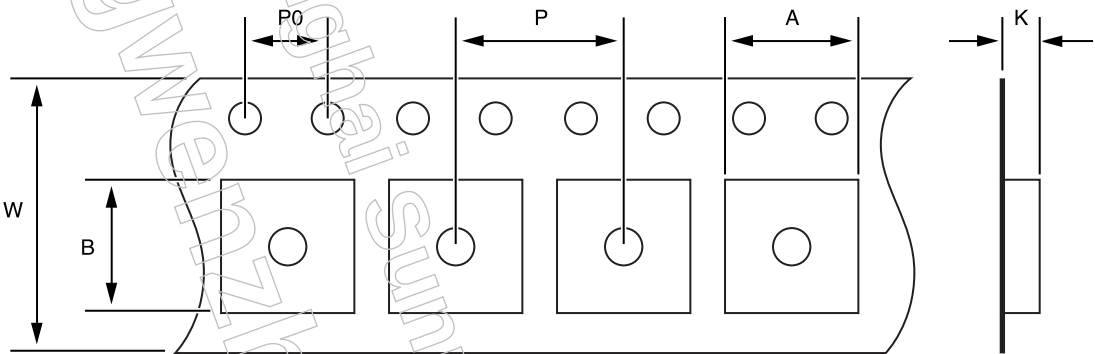


Table 79-4 Auth 3.0 CP packing tape dimensions (XDFN)

Symbol	Minimum	Nominal	Maximum
P0	3.90 mm	4.00 mm	4.10 mm
P	3.90 mm	4.00 mm	4.10 mm
A	2.15 mm	2.20 mm	2.25 mm
B	3.15 mm	3.20 mm	3.25 mm
K	0.48 mm	0.53 mm	0.58 mm
W	11.90 mm	12.00 mm	12.30 mm

Table 79-5 Auth 3.0 CP packing tape dimensions (WLCSP)

Symbol	Minimum	Nominal	Maximum
P0	3.90 mm	4.00 mm	4.10 mm
P	1.95 mm	2.00 mm	2.05 mm
A	1.24 mm	1.29 mm	1.34 mm
B	1.54 mm	1.59 mm	1.64 mm
K	0.44 mm	0.49 mm	0.54 mm
W	7.90 mm	8.00 mm	8.30 mm

Figure 79-5 Auth 3.0 CP packing reel (XDFN and WLCSP)

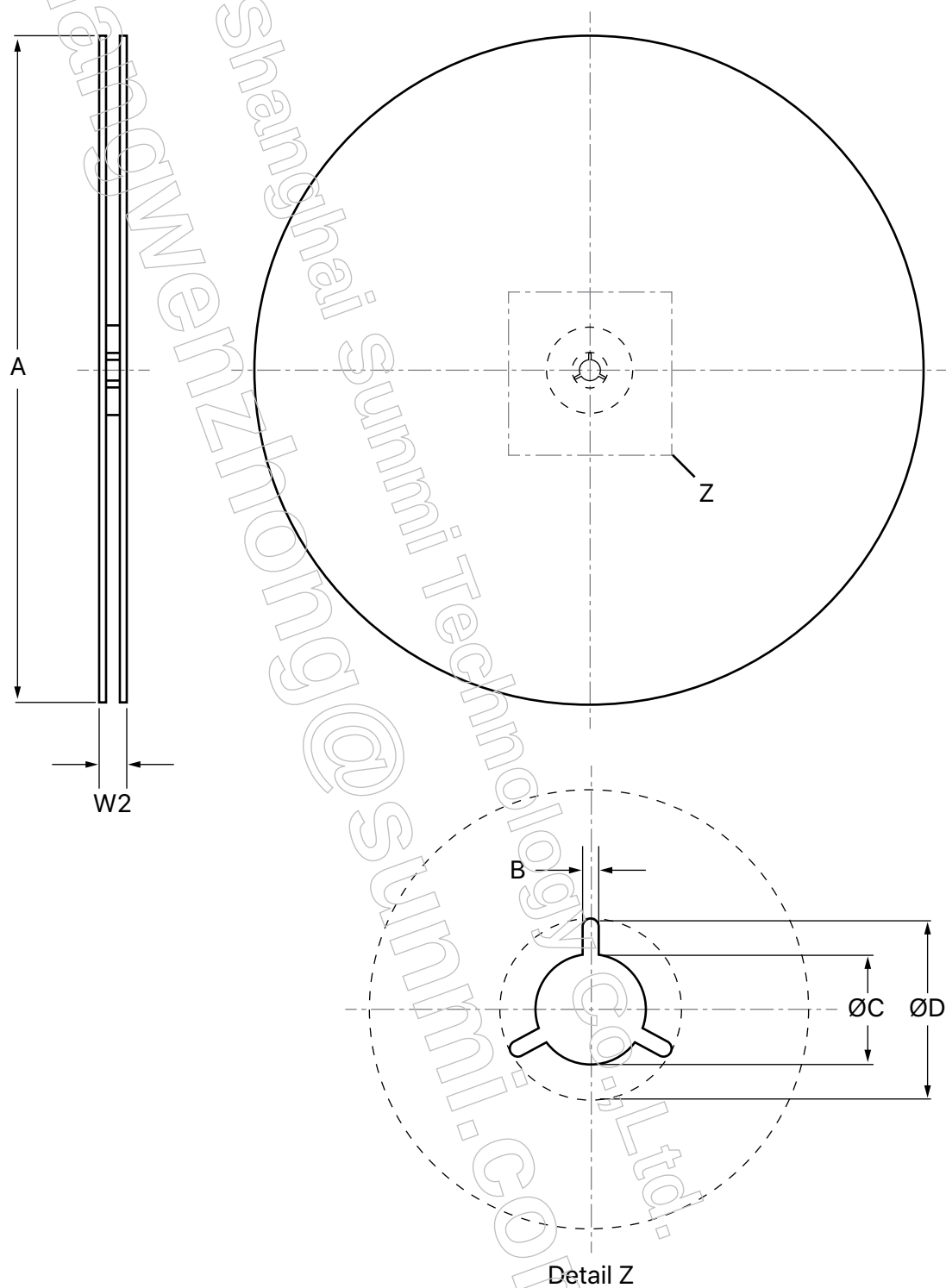


Table 79-6 Auth 3.0 CP packing reel dimensions (XDFN)

Symbol	Minimum	Nominal	Maximum
A	328.0 mm	330.0 mm	332.0 mm
W2			18.4 mm
B	1.5 mm	2.0 mm	2.5 mm
ØC	12.8 mm	13.0 mm	13.5 mm
ØD	20.2 mm		

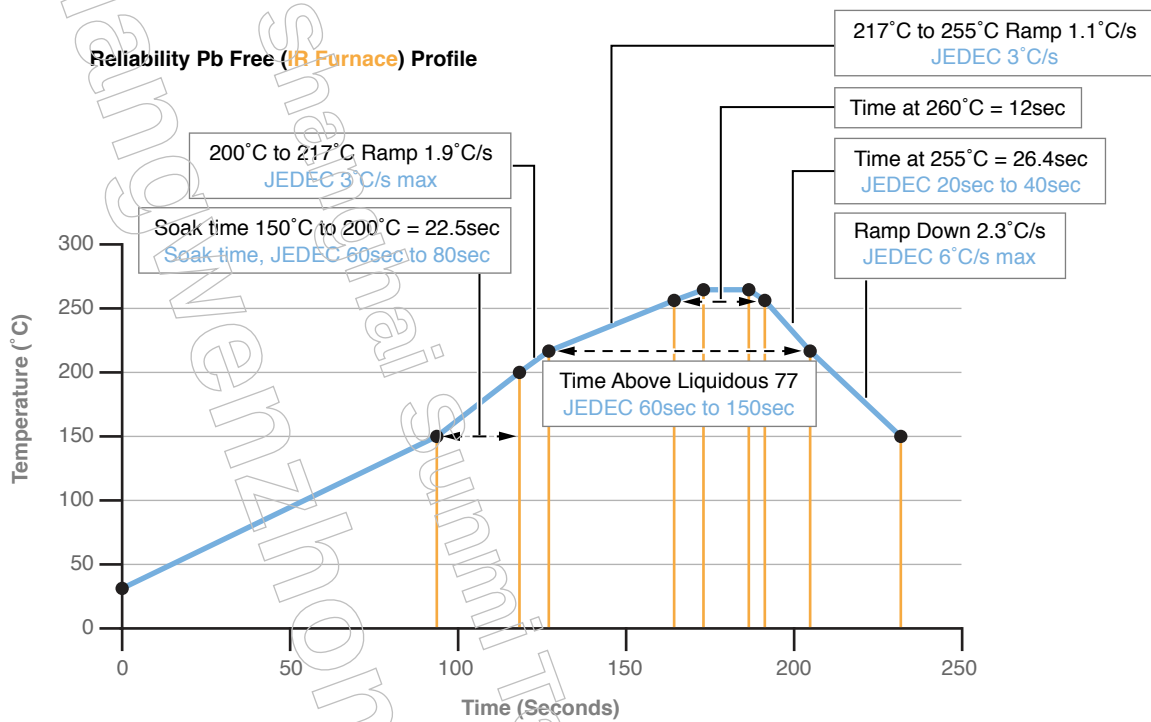
Table 79-7 Auth 3.0 CP packing reel dimensions (WLCSP)

Symbol	Minimum	Nominal	Maximum
A	328.0 mm	330.0 mm	332.0 mm
W2			14.4 mm
B	1.5 mm	2.0 mm	2.5 mm
ØC	12.8 mm	13.0 mm	13.5 mm
ØD	20.2 mm		

79.2.4 Handling

The Auth 3.0 CP package is Pb-free and is designed to withstand a standard reflow profile with a peak reflow temperature below 260 °C. See [Figure 79-6](#) (page 703) for the reflow profile of the Auth 3.0 CP (XDFN and WLCSP).

Figure 79-6 Auth 3.0 CP reflow profile (XDFN and WLCSP)



79.3 Pad Assignments

79.3.1 XDFN Package

Figure 79-7 Auth 3.0 CP, top view (XDFN)

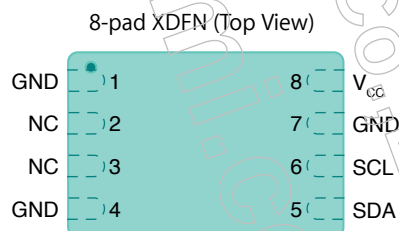


Table 79-8 Auth 3.0 CP signals (XDFN)

Signal	Pad	I/O	Description
GND	1		Ground
NC	2-3		Shall not be connected.
GND	4		Ground
SDA	5	I/O	I ² C data
SCL	6	I	I ² C clock
GND	7	I	Ground
V _{CC}	8		Supply voltage

Pads 1, 4, and 7 are internally connected to each other. The ground terminal of the bypass capacitor should be as close as possible to any one of these three pads.

79.3.2 WLCSP Package

Figure 79-8 Auth 3.0 CP, top view (WLCSP)

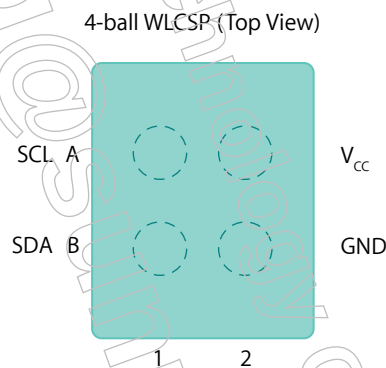


Table 79-9 Auth 3.0 CP signals (WLCSP)

Signal	Pad	I/O	Description
SCL	1A	I	I ² C clock
SDA	1B	I/O	I ² C data
V _{CC}	2A		Supply voltage
GND	2B		Ground

79.4 Electrical

This section provides technical details and tolerances for the Auth 3.0 CP chip.

79.4.1 Maximum Environmental Conditions

[Table 79-10](#) (page 705) lists the Auth 3.0 CP's absolute maximum electrical and free-air temperature ranges. Stresses to the Auth 3.0 CP beyond the ranges may cause permanent damage. Exposure to either end of any range may affect the Auth 3.0 CP reliability.

Table 79-10 Auth 3.0 CP maximum electrical and temperature ranges

Condition	Maximum Range
Voltage applied at V_{CC} relative to V_{SS}	-0.3 V to +6.0 V
Voltage applied to any pad	-0.3 V to $V_{CC} + 0.3$ V
Storage temperature	-65 to +150 °C

79.4.2 Recommended Operating Conditions

Internal sensors may force the Auth 3.0 CP to its reset state if any of the conditions listed in [Table 79-11](#) (page 705) are exceeded. Attempting to operate the Auth 3.0 CP in this state is not recommended and may lead to the Auth 3.0 CP failure or unreliability.

Table 79-11 Auth 3.0 CP maximum electrical and temperature ranges

Condition	Recommended Range
Operating free-air temperature	-40 to +85 °C
Supply voltage	1.71 V to 3.6 V

79.4.3 ESD Characteristics

Table 79-12 ESD characteristics

Pad	Description	ESD Requirement	Voltage
V_{CC}	Supply voltage	CDM ± 1 kV, HBM ± 4 kV	+3.6 V
SDA	I ² C data	IEC 61000-4-2 ± 8 kV Contact Discharge	+8.0 V limited to 150 ms max over lifetime.
SCL	I ² C clock	CDM ± 1 kV, HBM ± 4 kV	+5.5 V

79.4.4 DC Characteristics

Table 79-13 DC characteristics

Item	Description	Minimum	Typical	Maximum
I_{CHAL}	Operating current while computing challenge	-	-	1.15 mA
I_{SLEEP}	Sleep current	-	100 nA	2000 nA
V_{OL}	Output Voltage Low, $I_{\text{OL}} = 8 \text{ mA}$	-	-	0.4 V
V_{OL}	Output Voltage Low, $I_{\text{OL}} < 500 \mu\text{A}$	-	-	0.2 V
C_{SDA}	Input capacitance of SDA	-	60 pF	-
C_{SCL}	Input capacitance of SCL	-	3 pF	-
V_{CC}	Supply voltage	1.71 V	-	3.6 V
I_{IH}	Input Current ($V_{\text{IH}} = 0 \text{ V}, 3.6 \text{ V}$)	-1 μA	-	1 μA
V_{IH}	Input Voltage High	$0.7 \text{ V} \times V_{\text{CC}}$	-	3.9 V
V_{IL}	Input Voltage Low	-0.3 V	-	$0.3 \text{ V} \times V_{\text{CC}}$

79.4.5 AC Characteristics

This section is applicable over recommended operating range from $T_{\text{A}} = -40^\circ\text{C}$ to $+85^\circ\text{C}$, $V_{\text{CC}} = +1.71 \text{ V}$ to $+3.6 \text{ V}$.

Table 79-14 AC characteristics

Item	Description	Minimum	Typical	Maximum
t_{WATCHDOG}	Watchdog Timeout to drop into sleep	3.2 s	4 s	4.8 s
t_{DIT}	Auth 3.0 CP Initialization Time from power applied or wake from sleep	-	-	1 ms
t_{AUTH}	Authentication Time	-	-	500 ms
t_{CERT}	Certificate Generation	-	-	10 ms
t_{READ}	EEPROM Reads	-	-	1.5 ms
t_{WRITE}	EEPROM Writes	-	-	20 ms
t_{SN}	Serial Number Formatting	-	-	1 ms

Item	Description	Minimum	Typical	Maximum
t_{CMD}	All Other Commands	-	-	800 μs
f_{SCK}	SCL Clock Frequency	0 kHz	-	400 kHz
t_{HIGH}	SCK High Time	0.6 μs	-	-
t_{LOW}	SCK Low Time	1.2 μs	-	-
$t_{\text{SU.STA}}$	Start Setup Time	0.6 μs	-	-
$t_{\text{HD.STA}}$	Start Hold Time	0.6 μs	-	-
$t_{\text{SU.STO}}$	Stop Setup Time	0.6 μs	-	-
$t_{\text{SU.DAT}}$	Data In Setup Time	100 ns	-	-
$t_{\text{HD.DAT}}$	Data In Hold Time	0 ns	-	-
t_{R}	Input Rise Time from $0.1 \times V_{\text{CC}}$ to $0.9 \times V_{\text{CC}}$	-	-	1 μs
t_{F}	Input Fall Time from $0.9 \times V_{\text{CC}}$ to $0.1 \times V_{\text{CC}}$	-	-	1 μs
t_{AA}	Clock Low to Data Out Valid	-	-	0.9 μs
t_{DH}	Data Out Hold Time	50 ns	-	-
t_{BUF}	Time bus shall be free before a new transmission can start.	1.2 μs	-	-
t_{I}	Input Filter Glitch Suppression (SCL, SDA)	-	-	50 ns

Note:

Calculation time measured from the end of the trailing break after command input to the point in time when the Auth 3.0 CP begins transmitting the first bit of the response.

79.4.6 Sleep Mode

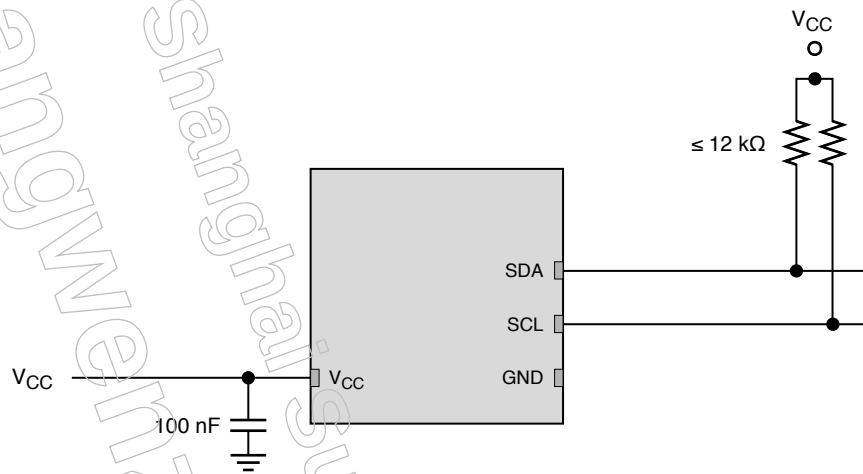
The Auth 3.0 CP conserves power by entering a sleep mode. It enters this mode automatically after the watchdog timer expires (t_{WATCHDOG}) or using the I²C [Sleep](#) (page 717) sequence.

The Auth 3.0 CP automatically wakes in response to any activity on SDA when in sleep mode. Auth 3.0 CP will NACK all data on SDA when waking from sleep.

79.4.7 Reference Design

Accessories shall implement the reference design as shown in [Figure 79-9](#) (page 708).

Figure 79-9 Auth 3.0 CP reference design



79.5 I²C Interface

The Auth 3.0 CP acts as a I²C device and uses a 7-bit address, see [Addressing](#) (page 709).

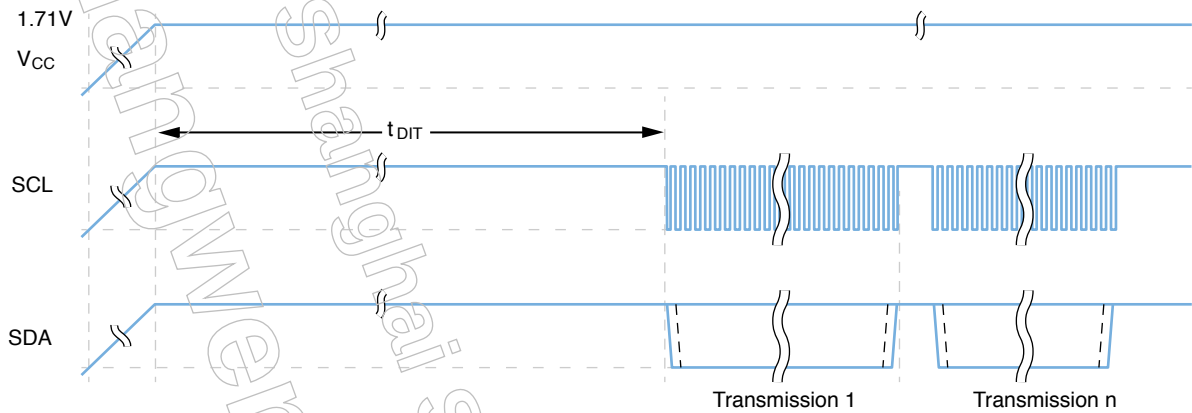
79.5.1 Setup

The I²C interface is started either by supplying power to the V_{CC} line or by any transition on the SDA line if the Auth 3.0 CP is in [Sleep Mode](#) (page 707).

The I²C output buffer is cleared on power-up and wake from sleep mode.

The Auth 3.0 CP requires a certain amount of time to read the I²C address when powered up. The first data transmission shall start no earlier than t_{DIT} , see [Table 79-14](#) (page 706), after the V_{CC} line goes high. See [Figure 79-10](#) (page 709) for an I²C timing diagram.

Figure 79-10 Auth 3.0 CP I²C power up timing



79.5.2 Hardware

The I²C Interface operates at 100 kHz (standard mode) or 400 kHz (full speed) and consists of two signals:

- SDA: Data
- SCL: Clock

Accessories shall connect pull-up resistors (12 k Ω maximum) from the same power supply as the Auth 3.0 CP to the SCL and SDA lines of the Auth 3.0 CP.

79.5.3 Addressing

The I²C device address for writing is shown in [Figure 79-11](#) (page 709) and the corresponding read address in [Figure 79-12](#) (page 709).

Figure 79-11 Auth 3.0 CP I²C device write address

A6	A5	A4	A3	A2	A1	A0	R/nW
0	0	1	0	0	0	0	0

Figure 79-12 Auth 3.0 CP I²C device read address

A6	A5	A4	A3	A2	A1	A0	R/nW
0	0	1	0	0	0	0	1

The I²C write address of the Auth 3.0 CP consists of the seven bits [A6:A0] followed by 0 for the R/nW bit. The I²C read address of the Auth 3.0 CP consists of the seven bits [A6:A0] followed by 1 for the R/nW bit. The write and read addresses of the Auth 3.0 CP are 0x20 and 0x21 respectively.

79.5.4 I²C Transaction Format

Table 79-15 Auth 3.0 CP read/write transaction data format

Field	Bytes	Description
I ² C Device Address + R/W bit	1	Shall be 0x20 for a write transaction or 0x21 for a read transaction.
Register	1	See Registers (page 712).
Payload	0-n	Payload data.

79.5.5 Writing to the Auth 3.0 CP

To write data to the Auth 3.0 CP, follow these steps:

1. Send the I²C start sequence.
2. Send the I²C write address of the Auth 3.0 CP.
3. Verify an ACK is received from the I²C device; if a NACK is received, wait 500 μ s and then loop back to Step 1.
4. Send the register address at which to begin writing.
5. Verify an ACK is received from the I²C device; if a NACK is received, wait 500 μ s and then loop back to Step 1.
6. Send the data bytes.
7. Send the I²C stop sequence.

The accessory shall send a I²C stop sequence after the last data byte of any write to ensure the Auth 3.0 CP will start performing the operation. The behavior of the Auth 3.0 CP is undefined if a I²C start sequence is sent prior to a I²C stop sequence.

79.5.6 Reading from the Auth 3.0 CP

To read data from the Auth 3.0 CP, follow these steps:

1. Send the I²C start sequence.
2. Send the I²C write address of the Auth 3.0 CP.

3. Verify an ACK is received from the I²C device; if a NACK is received, wait 500 μ s and then loop back to Step 1.
4. Send the register address at which to begin reading.
5. Verify an ACK is received from the I²C device; if a NACK is received, wait 500 μ s and then loop back to Step 1.
6. Send the I²C stop sequence.
7. Send the I²C start sequence.
8. Send the I²C read address of the Auth 3.0 CP.
9. Verify an ACK is received from the I²C device; if a NACK is received, wait 500 μ s and then loop back to Step 7.
10. Read the data bytes.
11. Send the I²C stop sequence.

If the read buffer on the accessory is limited and cannot read all the desired bytes prior to issuing the I²C stop sequence in Step 11, then the following sequence may be repeated as many times as necessary to retrieve all remaining bytes:

1. Send the I²C start sequence.
2. Send the I²C read address of the Auth 3.0 CP.
3. Verify an ACK is received from the I²C device; if a NACK is received, wait 500 μ s and then loop back to Step 6.
4. Read the data bytes.
5. Send the I²C stop sequence.

After all bytes in the register have been transmitted to the accessory, the Auth 3.0 CP will return 0x00 in response to all further reads until a new register number is selected.

79.5.7 Mutual Exclusivity

If the accessory supports authentication for multiple devices, transports, or features, each challenge generation process shall be mutually exclusive. The accessory shall not retrieve [Accessory Certificate Data](#) (page 716), write new [Challenge Data](#) (page 715), or start a new challenge response computation using [Authentication Control and Status](#) (page 713) until the previous operation is complete and the [Challenge Response Data](#) (page 715) has been read.

79.5.8 Registers

79.5.8.1 Device Version

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Device Version	0x00	0	uint8	0x07	1	Read-only

Reading from this register will return 0x07.

79.5.8.2 Authentication Revision

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Revision	0x01	0	uint8	0x01	1	Read-only

Reading from this register will return 0x01.

79.5.8.3 Authentication Protocol Major Version

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Protocol Major Version	0x02	0	uint8	0x03	1	Read-only

Reading from this register will return 0x03.

79.5.8.4 Authentication Protocol Minor Version

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Protocol Minor Version	0x03	0	uint8	0x00	1	Read-only

Reading from this register will return 0x00.

79.5.8.5 Device ID

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Device ID	0x04	0	uint8	00 00 03 00	4	Read-only

Reading from this register will return 00 00 03 00.

79.5.8.6 Error Code

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Error Code	0x05	0	uint8	0x00	1	Read-only

Table 79-16 Error Code register parameters

Bit	Name	Description
7-0	Error Code	See Table 79-17 (page 713).

Table 79-17 Error Code values

Value	Description
0x00	No error.
0x01	Reserved
0x02	Invalid register specified or register is read-only.
0x03-0x04	Reserved
0x05	Sequence error, command out of sequence.
0x06	Internal process error during challenge response generation.
0x07-0xFF	Reserved

Reading from this register will clear it.

Attempts to access an undefined register may result in an error code of 0x02.

79.5.8.7 Authentication Control and Status

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Authentication Control and Status	0x10	1	uint8	0x00	1	Read/Write

Reading from this register provides the status of the most recently requested Auth 3.0 CP process, as shown in [Figure 79-13](#) (page 714) and [Table 79-18](#) (page 714).

Writing to this register controls the start of the Auth 3.0 CP processes, as shown in [Figure 79-14](#) (page 714) and [Table 79-21](#) (page 715).

Figure 79-13 Auth 3.0 CP Authentication Control and Status register, read-only bits

7	6	5	4	3	2	1	0
Error Set	Procedure Results			0	0	0	0

Table 79-18 Authentication Control and Status read-only register parameters

Bit	Name	Description
7	Error Set	See Table 79-19 (page 714).
6-4	Procedure Results	See Table 79-20 (page 714).
3-0		Reserved

Table 79-19 Error Set values

Value	Description
0	The Error Code register contains a value of zero.
1	The Error Code register contains a non-zero value. Both this bit and the Error Code register contents are cleared after the Error Code register is next read or on the next power-on.

Table 79-20 Procedure Results values

Value	Description
0	Most recent challenge response generation process did not produce valid results or no challenge response generation process has been requested.
1	Accessory challenge response successfully generated.
2-7	Reserved

Figure 79-14 Auth 3.0 CP Authentication Control and Status register, write-only bits

7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	Proc Control

Table 79-21 Authentication Control and Status write-only register parameters

Bit	Name	Description
7-1		Reserved
0	Procedure Control	Set to start new challenge response generation process.

Writing 0x01 to this register initiates challenge response computation.

79.5.8.8 Challenge Response Data Length

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Response Data Length	0x11	1	uint16	0	2	Read-only

This register holds the length in bytes of the results of the most recent challenge response generation process. If a completed challenge response is stored in the output buffer of the Auth 3.0 CP, this register will return a value of 64. In all other cases it will return a value of zero.

79.5.8.9 Challenge Response Data

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Response Data	0x12	1	uint8	Undefined	64	Read-only

If the Auth 3.0 CP has computed a challenge response, the challenge response itself will be returned in this register. If there is no valid challenge response stored in the Auth 3.0 CP SRAM, the interface will return 0x00 for all bytes.

79.5.8.10 Challenge Data Length

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Data Length	0x20	2	uint16	0	2	Read-only

This register holds the length in bytes of the current challenge stored within the Auth 3.0 CP. If a challenge is valid within the Auth 3.0 CP it will return a value of 32, otherwise, it will return a value of zero on a read.

79.5.8.11 Challenge Data

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Challenge Data	0x21	2	uint8	Undefined	32	Read/Write

This register holds the current challenge data. If a valid challenge is stored within the Auth 3.0 CP, its value may be read using this register, otherwise reads of this register return all zeros.

This challenge data shall be written to the Auth 3.0 CP by the accessory and shall always be 32 bytes in length. If any number of bytes other than 32 is sent to the Auth 3.0 CP, the challenge register will be cleared and an error returned.

79.5.8.12 Accessory Certificate Data Length

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Accessory Certificate Data Length	0x30	3	uint16	607-609	2	Read-only

This register holds the length of the X.509 certificate that is stored within the Auth 3.0 CP. The length of a certificate varies between 607 and 609, inclusive.

79.5.8.13 Accessory Certificate Data

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Accessory Certificate Data 1	0x31	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 2	0x32	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 3	0x33	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 4	0x34	3	uint8	Certificate	128	Read-only
Accessory Certificate Data 5	0x35	3	uint8	Certificate	128	Read-only

These registers hold the binary X.509 certificate stored in Auth 3.0 CP. The accessory certificate may be read from the Auth 3.0 CP in 128-byte pages starting at any of Accessory Certificate Data addresses, or it may be read in a continuous stream starting at Accessory Certificate Data 1. The Accessory Certificate Data Length value can be read to determine the exact number of bytes in the Accessory Certificate Data.

The accessory shall wait at least t_{CERT} for the Auth 3.0 CP to create the certificate from the stored elements whenever it is requested. This means if any register within block 3 is selected using a zero length write, the subsequent read operation shall not be issued for t_{CERT} , even if the only intention is to read the length register.

79.5.8.14 Self-Test Status

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Self-Test Status	0x40	4	uint8	0x00	1	Read-only

This register reports the results of the X.509 certificate and private key tests, as shown in [Figure 79-15](#) (page 717) and [Table 79-22](#) (page 717).

Figure 79-15 Auth 3.0 CP Self-Test Status register



Table 79-22 Self-Test Status register parameters

Bit	Name	Description
7	X.509 Certificate	See Table 79-23 (page 717).
6	Private key	See Table 79-23 (page 717).
5-0		Reserved

Table 79-23 Self-Test Status values

Value	Description
0	Error
1	Pass

Note:

The X.509 and private key tests only verify these elements are present in flash memory; no authentication is performed.

79.5.8.15 Device Certificate Serial Number

Name	Address	Block	Type	Power-Up Value	Bytes	Access
Device Certificate Serial Number	0x4E	4	uint8	Certificate	32	Read-only

This register returns the certificate serial number as an ASCII string. Each byte in the serial number stored within the device will be returned as two ASCII characters in the range of 0-9, A-F.

79.5.9 Sleep

The sequence to enter sleep mode for the Auth 3.0 CP is:

Table Auth 3.0 CP sleep
79-24

Operation	Value
Start Condition	
Address + Write	0x34
Register Address	0x10
Value	0x60
Value	0xE0
Value	0x04
Value	0x01
Value	0x20
Value	0x1D
Value	0xEE
Value	0x00
Stop Condition	
Wait for 10 μ s.	
Start Condition	
Address + Write	0x34
Register Address	0x10
Value	0x60
Stop Condition	